

Intelligent Travel Concierge Assistant

Industry / Sector: Travel & Hospitality

Company Overview

GrandStay Hospitality Group (GSH)

Headquarters: Alberta, Canada

GrandStay Hospitality Group (GSH) was founded in 1955 as a small, family-owned luxury hotel in the Mid-Atlantic region. Over the decades, GSH has grown into one of North America's most recognizable upscale hotel chains, operating over 8,000 properties across 130 countries under multiple brands ranging from business hotels to destination resorts.

The company's success has been built on personalized service, loyalty programs, and operational excellence. GSH was among the first in the industry to implement mobile check-in, keyless room access, and AI-driven revenue management systems.

Today, its customer base includes both business and leisure travelers, with a growing focus on digital engagement and sustainability-driven tourism. GSH differentiates itself through its "Stay Your Way" philosophy, an experience-centered approach that prioritizes seamless guest interactions before, during, and after stays.

What sets GSH apart is its commitment to digital transformation in hospitality from predictive maintenance using IoT data to AI-based guest personalization. However, as customer expectations for instant, intelligent service continue to rise, the company recognized an urgent need to modernize its guest interaction systems.

Business Challenge

Despite GSH's strong reputation for customer service, the company faced an operational challenge that had grown increasingly costly and inefficient. Guests frequently contacted customer support to ask repetitive questions about:

- Hotel amenities (Wi-Fi, gym, parking, dining hours)
- Local attractions and transportation options
- Booking, cancellation, and loyalty program policies

Each of these inquiries required manual responses from call center staff or front desk teams, creating bottlenecks and inconsistent information delivery across regions.

Key Contributing Factors:

- High call and email volume (up to 60% of inquiries were repetitive in nature).
- Rising labor costs due to extended support hours across multiple time zones.
- Guest frustration with slow response times and inconsistent answers.
- Missed opportunities for cross-selling services (spa, tours, upgrades).

Business Impacts:

- Increased operational costs and staff fatigue.
- Declining Customer Satisfaction (CSAT) and Net Promoter Scores (NPS).
- Inhibited ability to deliver real-time, contextual assistance.

In a highly competitive hospitality market, where digital-first customer experiences are becoming the norm, GSH recognized that improving service speed and personalization wasn't just desirable — it was essential for retaining guests and sustaining loyalty.

Rationale for the Project

The Intelligent Travel Concierge Assistant initiative was conceived as part of GSH's broader "Digital Guest Experience Transformation" strategy. The project's goal is to deploy an AI-powered, retrieval-augmented concierge capable of answering guest queries instantly and accurately — both during and before their stay. This approach aligns with a growing industry trend: the integration of Large Language Models (LLMs) in hospitality to enhance guest engagement and automate low-value, repetitive interactions.

Industry Examples:

- Hilton has experimented with AI chatbots that integrate with WhatsApp and Google Assistant.
- Accor uses AI tools to personalize guest recommendations based on historical preferences.
- Expedia Group leverages GPT-powered systems to offer contextual travel planning assistance.

These examples demonstrate that AI-driven guest communication is becoming a competitive differentiator.

Top Five Reasons for Initiating the Project:

Enhance Guest Experience

Provide instant, personalized, and accurate responses 24/7.

Reduce Operational Costs

Automate repetitive tasks, freeing staff to focus on complex guest needs.

Consistency & Compliance

Ensure all guests receive standardized, policy-compliant answers.

Upselling Opportunities

Integrate intelligent recommendations (e.g., dining, spa, local tours).

Strategic Differentiation

Reinforce GSH's market leadership through innovation and digital excellence.

By addressing these priorities, GSH aims to achieve measurable improvements in guest satisfaction, loyalty, and cost efficiency.

Project Objectives

The Intelligent Travel Concierge Assistant aims to achieve the following objectives:

1 Deploy a Generative AI Concierge

Create an AI assistant using LangChain and OpenAI's/ Gemini/ DeepSeek GPT models capable of responding conversationally to guest inquiries.

2 Enable Contextual Information Retrieval

Integrate ChromaDB as a vector database to store and retrieve internal documents (amenities info, policies, guides) using semantic search.

3 Deliver an Interactive Guest Interface

Build a user-friendly, web-based Streamlit application accessible via the GSH website and mobile app.

4 Automate Knowledge Updates

Implement an ingestion pipeline that periodically updates embeddings from new documents (e.g., policy changes or new attractions).

5 Measure Impact Through KPIs

Reduce customer support inquiries by 40%. Improve average response time by 25%. Achieve 15% higher CSAT scores.



Comprehensive Dataset

For this project, GSH will not rely on traditional structured datasets but instead utilize document-based and semi-structured data sources, such as:

- Internal policy manuals and guest FAQs (PDFs, Word files).
- Amenity and service guides (text documents).
- Past customer chat transcripts for training the assistant on tone and phrasing.

These resources will be pre-processed into text chunks, embedded using OpenAI's embedding model, and stored in ChromaDB for retrieval.

DS Technology Stack

Component	Technology / Tool	Purpose
Language Model	Deepseek	Generate conversational responses
Framework	LangChain	Build the RAG pipeline and conversation flow
Vector Database	ChromaDB	Store and query embeddings of hotel documents
Frontend Interface	Streamlit	Build an interactive chat-based UI
Data Preprocessing	Python, PyPDF, Pandas	Extract and clean data from internal documents
Hosting / Integration	AWS / Azure	Deploy the application at scale

PM Technology Stack

Component	Technology / Tool	Purpose
Task Management	Jira	Task breakdown, sprint planning, issue tracking, KPI monitoring, and progress tracking
Documentation & Collaboration	Confluence	Project documentation, progress reporting, knowledge sharing, risk management
Visual Communication	Draw.io	Visualize architecture, workflows, system interactions, and process flow diagrams

Data Science Project Scope

The project will follow a structured, end-to-end data science workflow:

Problem Definition & Scoping

Define core user queries and use cases (e.g., amenities, policies, attractions). Identify key business KPIs (response time, ticket reduction, CSAT).

Data Acquisition & Preparation

Collect hotel documents, FAQs, and policy PDFs. Clean and segment text into meaningful chunks. Generate embeddings using HuggingFace embedding model (all-MiniLM-L6-v2).

Vector Database Setup

Create a ChromaDB instance. Store all embeddings and relevant metadata (e.g., category, region, language).

LangChain Pipeline Development

Implement Retrieval-Augmented Generation (RAG) pipeline: User query → Query embedding → Retrieve top-k documents from ChromaDB → Pass to GPT model for contextual answer. Define custom LangChain chains for response generation, summarization, and context memory.



Implementation & Deployment

Frontend Development

Build an intuitive Streamlit-based chatbot interface with hotel branding. Enable conversation memory and live context display.

Testing & Evaluation

Conduct user testing with real guest queries. Measure response accuracy, latency, and guest satisfaction.

Deployment & Monitoring

Deploy via AWS EC2 or Azure Web App. Implement a continuous feedback loop — new queries feed into retraining and document updates.

Impact Assessment

Monitor KPIs: Reduction in call center tickets. Guest satisfaction trends. Accuracy and latency of responses.

Daily Deliverables for Project Managers (PM) – 3 Weeks

Day	Task	Tool(s)	Deliverables
Day 1-2	Project scope finalization, roles and responsibilities, and Jira setup	Jira, Confluence	Initial project plan, Jira board set-up, Confluence page for project documentation
Day 3-4	Collect necessary documents and define initial tasks for backend and frontend	Confluence, Jira	Task assignments for data collection, document segmentation, and initial backend setup
Day 5-6	Team meeting for progress check and task updates	Jira, Confluence	Progress updates, revised task deadlines, Jira updates for task status
Day 7-8	Conduct first sprint review: assess progress on embeddings, LangChain pipeline, and UI	Jira, Confluence	Sprint review document in Confluence, Jira sprint progress report
Day 9-13	Run second sprint	Jira, Confluence	Test report, Jira updates on any issues or bugs found
Day 14	Sprint retrospective and re-assessment of task timelines	Jira, Confluence	Retrospective meeting minutes in Confluence, Jira task adjustments
Day 15-16	Focus on final testing, deployment preparations	Draw.io, Jira	Final system architecture diagram in Draw.io, deployment checklist in Jira
Day 17-18	Monitor deployment and post-launch performance metrics	Jira, Confluence	Post-launch report, Jira task updates for any issues during deployment
Day 19-20	Ensure the system is performing as expected, start monitoring KPIs	Jira, Confluence	KPI tracking document in Confluence, Jira updates for issues
Day 21	Review project impact and create final documentation for launch	Confluence, Jira	Final project report, Confluence document with post-launch analysis, Jira task closure